Quadratic Kernel Mapping and Kernel Trick

$$u = (u_1, u_2) \quad v = (v_1, v_2)$$

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_1 u_2 v_1 v_2 = \phi(u) \cdot \phi(v)$$

$$\phi(u) \cdot \phi(v) = [1, u_1, u_2, u_1 u_2] \cdot [1, v_1, v_2, v_1 v_2] = 1 + u_1 v_1 + u_2 v_2 + u_1 u_2 v_1 v_2$$

$$\text{where } \phi(x) = (x_1, x_2, x_1 x_2) \longrightarrow \mathbb{R}^3$$